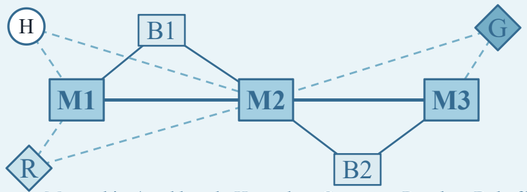
(a) Heterogeneous resource representation



M: machine/workbench H: worker G: gantry R: robot B: buffer

Graph G
M^G, E_lap

Node-aligned history

X: 30 × 38 × 27

6 groups

Grouped Linear + GELU
Mean aggregation

Time + Laplacian
+ node embeddings

$H^{(0)}$

(b) Process-informed rules and clustered temporal-pattern construction

Operational evidence

Queue | Stall / shortage | Logistics

Resource-specific rules + persistence

Raw

Sustained

Event / state targets

Cause targets

Supervision

History-derived conditioning

Rule-based regime IDs → **Regime embedding**

Local + neighbour cues → **Precursor MLP**

Clustered temporal-pattern construction

Activity patches (length 3) → *k-means* → 16 prototypes P

PHGST-DP

Encoder block × 5

LayerNorm

Temporal (4 heads) | **Geographic** (2 heads, M^G) | **Semantic** (2 heads, M^s)

Concat + Linear

residual

LayerNorm

Feed-forward network

residual

$H(t)$

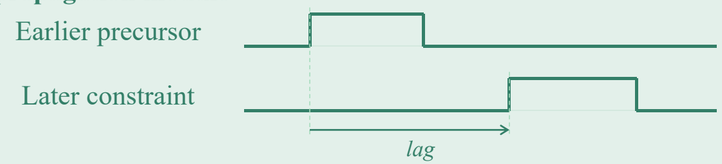
Regime + precursor fusion (b) | **Skip + order-progress fusion** | **Graph pooling + node evidence**

Event heads | **State / remaining time decoders** | **Process-cause classifier**

Training loss

Event decoding and report assembly

(c) Prototype-conditioned delay-aware propagation module



Pattern matching

Queue query patch

Activity prototypes P (b)

Q^P (38 × 20) | K^P (16 × 20) | V^P (16 × 20)

$\alpha^P = \mathrm{softmax}(Q^P K^{PT} / \sqrt{d_h})$

Pattern weights 38 × 16

Pattern context C^P 38 × 20

Geographic key update

$\tilde{K}^G = K^G + C^P$

Base key K^G + Context C^P → **Updated key $\tilde{K}^G$**

Q^G, V^G → $\mathrm{softmax}(Q^G \tilde{K}^{GT} / \sqrt{d_h} + M^G) V^G$

Structured bottleneck report

